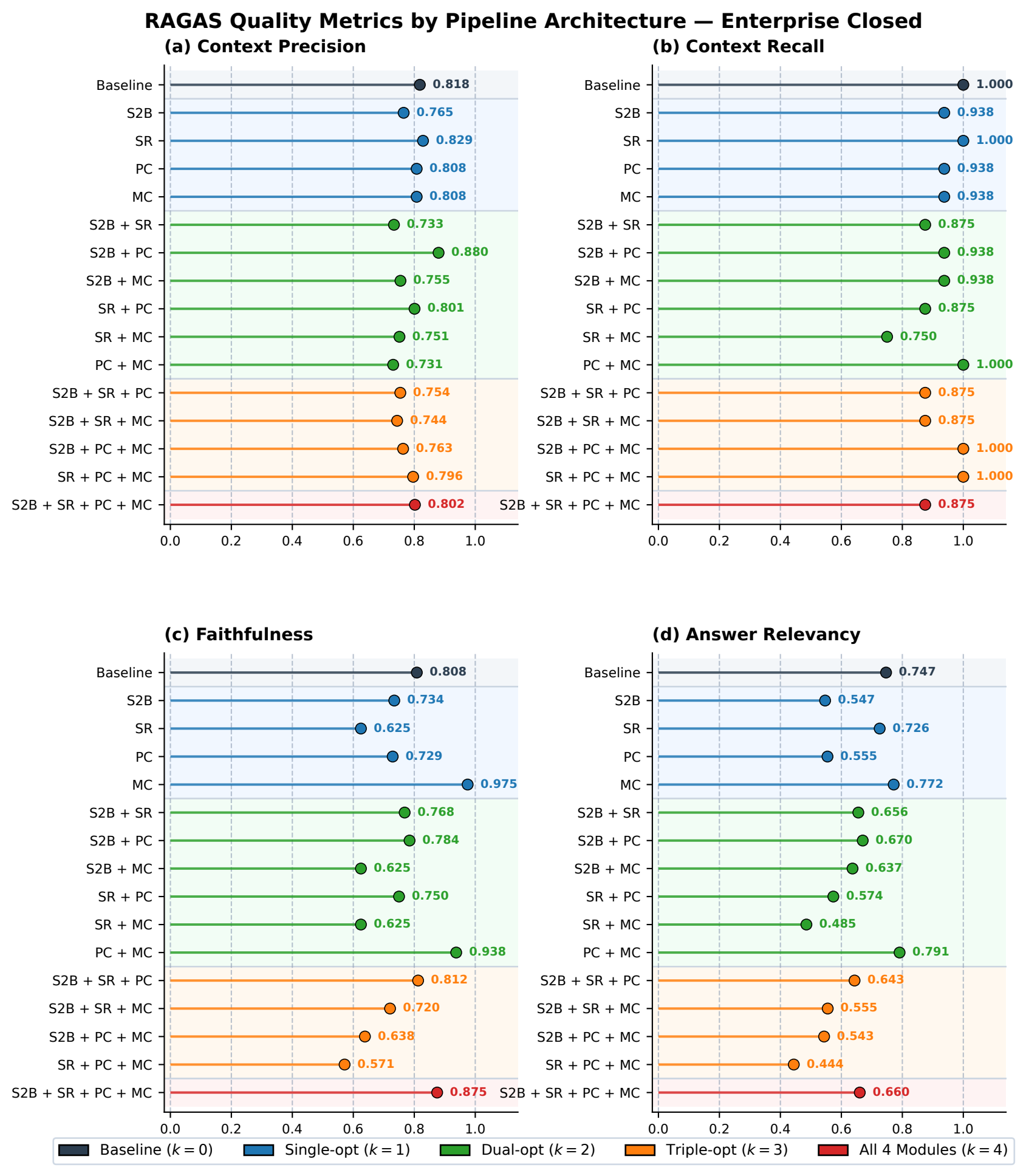
RAGAS Quality Metrics by Pipeline Architecture — Enterprise Closed
(a) Context Precision
Baseline 0.818
S2B 0.765
SR 0.829
PC 0.808
MC 0.808
S2B + SR 0.733
S2B + PC 0.880
S2B + MC 0.755
SR + PC 0.801
SR + MC 0.751
PC + MC 0.731
S2B + SR + PC 0.754
S2B + SR + MC 0.744
S2B + PC + MC 0.763
SR + PC + MC 0.796
S2B + SR + PC + MC 0.802
0.0 0.2 0.4 0.6 0.8 1.0
(b) Context Recall
Baseline 1.000
S2B 0.938
SR 1.000
PC 0.938
MC 0.938
S2B + SR 0.875
S2B + PC 0.938
S2B + MC 0.938
SR + PC 0.875
SR + MC 0.750
PC + MC 1.000
S2B + SR + PC 0.875
S2B + SR + MC 0.875
S2B + PC + MC 1.000
SR + PC + MC 1.000
S2B + SR + PC + MC 0.875
0.0 0.2 0.4 0.6 0.8 1.0
(c) Faithfulness
Baseline 0.808
S2B 0.734
SR 0.625
PC 0.729
MC 0.975
S2B + SR 0.768
S2B + PC 0.784
S2B + MC 0.625
SR + PC 0.750
SR + MC 0.625
PC + MC 0.938
S2B + SR + PC 0.812
S2B + SR + MC 0.720
S2B + PC + MC 0.638
SR + PC + MC 0.571
S2B + SR + PC + MC 0.875
0.0 0.2 0.4 0.6 0.8 1.0
(d) Answer Relevancy
Baseline 0.747
S2B 0.547
SR 0.726
PC 0.555
MC 0.772
S2B + SR 0.656
S2B + PC 0.670
S2B + MC 0.637
SR + PC 0.574
SR + MC 0.485
PC + MC 0.791
S2B + SR + PC 0.643
S2B + SR + MC 0.555
S2B + PC + MC 0.543
SR + PC + MC 0.444
S2B + SR + PC + MC 0.660
0.0 0.2 0.4 0.6 0.8 1.0
Baseline (k = 0)
Single-opt (k = 1)
Dual-opt (k = 2)
Triple-opt (k = 3)
All 4 Modules (k = 4)